

4(a). Mat growth features

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Introduction

Because sandy siliciclastic subaqueous environments are not typically characterized by mineral precipitation, their microbially mediated sedimentary structures have more subtle vertical relief when compared with microbial structures produced in mineral-precipitating carbonate environments. For a long time sedimentologists and palaeobiologists only worked on the more obvious stromatolites and associated carbonate structures, and microbial mat features from siliciclastic rocks were overlooked. But, in the past decade close examination of siliciclastic sedimentary rocks, particularly those from Precambrian and early Phanerozoic marine strata, has revealed a plethora of structures, largely formed due to growth of microbial mats at the sediment-water interface or the interaction of physical processes with biofilm- or mat-bound sediment surfaces.

There are numerous relevant references to these structures, which are listed in this book and referred to in many of the other chapters. We draw readers' attention to certain critical references: Hagadorn and Bottjer (1997), Hagadorn et al. (1999), Gerdes et al. (2000a), Noffke et al. (2001a) and Schieber (2004).

Definition and genesis of mat growth features

When a microbially-bound sandy surface is subjected to current and wave action, the surface may tear and break apart into cohesive fragments that can range from having a pliable consistency to being relatively rigid (Figs. 4(a)-1, -2, -3A). A variety of evidence can indicate the original pliable nature of such clasts, including deformation of adjacent clasts (Figs. 4(a)-1, -2). Sand chips indicate the possible influence of both microbial binding and mineral precipitation (Fig. 4(a)-3A).

Increased cohesiveness of sandy substrates imparted by the presence of microbial mat binding is also indicated by the stabilization of ripples on seafloors that are overlain by a subsequent generation of ripples (Fig. 4(a)-3B to -3D). In these cases the underlying rippled surface can show no evidence for erosion from deposition of the overlying rippled sand, indicating that the underlying surface was stabilized by microbial binding. Evidence for loading of the underlying rippled surface can also be used to demonstrate increased cohesiveness due to presence of a microbial mat (Figs. 4(a)-3C, -3D).

A variety of patterned bedding plane structures have been termed wrinkle structures and include such features as wrinkle marks (also known as 'runzelmarken'), 'old elephant skin' (OES), and 'Kinneyia' (Figs. 4(a)-4 to -8). Wrinkle structures (Figs. 4(a)-4A to -4E) may have multiple origins. When they occur at sandstone bed interfaces in which mud is demonstrably absent, the corrugated surface of the underlying sand bed likely reflects the original surface topography of a microbial mat, and/or micro-scale loading of these microbially-bound sediments (see also,

In: *Atlas of microbial mat features preserved within the clastic rock record*, Schieber, J., Bose, P.K., Eriksson, P.G., Banerjee, S., Sarkar, S., Altermann, W., and Catuneau, O., (Eds.) J. Schieber et al. (Eds.), Elsevier, p. 53-71. (2007)

discussion in Chapter 6(a)). For example, ‘old elephant skin’ (Figs. 4(a)-5 to -7) is caused by small-scale loading due to deposition of a thin sand bed upon a microbially-bound sandy seafloor. ‘Kinneyia’ (Fig. 4(a)-4F) is a surface of reticulated flat-topped ridges that are interpreted to form by gas bubble buildup beneath a biofilm, which inhibits cementation. When such features form in a series of successive beds, the phenomenon of exfoliating sand laminae may present itself in outcrop (Fig. 4(a)-8).

In sandy siliciclastic settings, significant topography may build up on the seafloor in the form of domal sand buildups or ‘sand stromatolites’ (Figs. 4(a)-9 to -11). Patterns in these structures can include clusters with larger domes in the middle surrounded by smaller domes (Figs. 4(a)-9, -10A, -10C, -10D), while other associations are not clustered and show no preferential size distribution (Figs. 4(a)-10B, -11). Clustered and non-clustered associations of domes can occur on the same bedding surface (Fig. 4(a)-10). The low synoptic relief of these domal sand buildups, in contrast with stromatolites, is likely due to the lack of carbonate precipitation in sandy siliciclastic environments, which aids in buildup of microbialites with greater vertical relief in carbonate settings.

In environments where sand stromatolites have formed, current action may transport sand across the seafloor, leading to the development of sand shadow structures (Fig. 4(a)-12). Ripples can also develop differentially on seafloors where the coverage by microbial mats was patchy, either due to original growth patterns or because mats were partially ripped-up by waves and/or currents (Figs. 4(a)-13, -14). Ripples with cracked crests are also found (Fig. 4(a)-15), most likely formed by fluid expulsion through the crests of the microbially-bound ripples. Such cracking of microbially-bound ripple crests can then lead to significant erosion of crests (Fig. 4(a)-16).

A variety of other features associated with microbial mat growth has also been described and is outlined in other sections of this book. Sand ‘roll-ups’ and petees also form during mat growth; in common with many of the microbial mat-related features in this book, these features may also be associated with genetic processes other than mat growth, and mat destruction-related examples are examined in section 4(c) of this chapter. Although we have made great progress over the past decade in identifying microbial mat growth features in sandy siliciclastic environments, we still have much to learn about what controls the morphology and distribution of microbial structures in siliciclastic environments – especially in ancient settings for which there are no modern comparative analogues.

Formal description of mat growth features

Spheroidal pliable sand clasts: Figs. 4(a)-1, -2, -3A

Name of structure: spheroidal pliable sand clasts.

Other terms used: ‘algal balls’, ‘sand balls’, ‘sand ooids’.

Description of structure: sphere-shaped clasts of sandstone showing no internal evidence for either clay or internal concentric lamination when examined through slabs, X-rays and thin sections. Clasts commonly deformed or deform one another; show no tendency towards sorting

into ripple troughs or (palaeo-)topographic depressions. May deform overlying or underlying sandstone beds, or be preserved as impressions in either of these bounding sandy beds.

Associated sedimentary structures: (1) ripple marks; (2) trace fossils.

Palaeoenvironment: intertidal to very shallow subtidal marine.

Ideas on genesis: growth of a mat which is subject to high energy wave and current action in shallow water (largely marine) settings can result in cohesive fragments of mat-bound sand which are then able to become rounded and pliable during their transport and deposition; deformation of these clasts and unlithified under- and overlying beds may then easily result. Spheroidal sandy fecal pellets can have similar shapes, sizes, and textures in Phanerozoic marginal marine settings; at present it is unknown how/if one can distinguish between such pellets and microbially-mediated sand spheroids.

Palimpsest ripples and loaded ripples: Figs. 4(a)-3B to -3D

Name of structure: palimpsest ripples, and loaded ripples.

Other terms used: none.

Description of structure: preservation of successive sets of ripples in sandy sediments, with crests of each set being relatively sharp and un-reworked. An absence of mud between such successive rippled sandy beds is essential to their identification; non-amalgamation of two rippled sandstone beds suggests the presence of a mat or biofilm providing organically-mediated sand stabilization. For loaded ripples, the rippled sandstone surface is marked by loading features related to a thin veneer of overlying sandstone.

Associated sedimentary structures: palimpsest and loaded ripples can occur together.

Palaeoenvironment: intertidal marine.

Ideas on genesis: the preservation of underlying ripples despite deposition of an overlying sandy, rippled bed argues for the presence of a microbial mat separating the two successive sandy beds, protecting the earlier ripples from reworking. For loaded ripples, the evidence for soft sediment deformation of the lower sandy bed and the lack of amalgamation of the two sandy beds argues for the presence of a mat to provide flexibility and cohesiveness of the lower, deformed bed.

Wrinkle structures: Figs. 4(a)-4A to -4E

Name of structure: wrinkle structures.

Other terms used: 'runzelmarken', wrinkle marks.

Description of structure: wrinkled sandstone or muddy sandstone surfaces, either as bed-tops or bed-soles, showing a reticulate pattern of small raised ridges.

Associated sedimentary structures: 'Kinneyia'.

Palaeoenvironment: intertidal to subtidal marine.

Ideas on genesis: wrinkle structures are thought to form due to either purely microbial processes or a combination of these with biofilm-mediated physical loading processes (see also, discussion in Chapter 6(a)). Wrinkle structures may reflect original topography of a microbial mat growing on a sandy substrate, or such microbially-bound sediment affected by micro-scale loading or shear stresses.

'Old elephant skin': Figs. 4(a)-5 to -7

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Name of structure: ‘old elephant skin’.

Other terms used: abbreviated as OES.

Description of structure: a surface corrugation of a sandstone bed which provides a quasi-polygonal texture in the rock record. Bed-top OES usually bears concave bulbous depressions and sharp convex crests, and bed-bottom OES bears convex-to-bulbous textures with sharp concave depressions.

Associated sedimentary structures: (1) oscillation ripples; (2) exfoliating sand laminae.

Palaeoenvironment: intertidal to subtidal marine.

Ideas on genesis: thought to reflect biofilms separating thin sandstone beds subject to soft sediment loading processes – the biofilm keeps the two beds separate and prevents their amalgamation, and also provides cohesiveness to the lower sandy bed to allow its deformation.

‘Kinneyia’: Fig. 4(a)-4F

Name of structure: ‘Kinneyia’.

Other terms used: flat-topped ‘runzelmarken’ or wrinkle marks.

Description of structure: irregular mosaic-like pattern of flat-topped ovate pits or grooves. Walls of pits or grooves are steep sided and occur on sandstone or muddy sandstone bed-tops or flat tops between truncated ripple crests.

Associated sedimentary structures: (1) wrinkle structures; (2) large ripple troughs.

Palaeoenvironment: intertidal to shallow marine.

Ideas on genesis: thought to form through buildup of gas bubbles in sandy sediments immediately underlying a biofilm or mat. Gas may be generated by decomposition of organic matter. Bubbles trapped on the underside of the biofilm may coalesce; sand between bubbles becomes cemented, and with subsequent decomposition of the overlying mat and compaction, overlying sediments infill the bubble pits.

Exfoliating sand laminae: Fig. 4(a)-8

Name of structure: exfoliating sand laminae.

Other terms used: none.

Description of structure: successive, very thin (≤ 1 mm) veneers of sandstone which spall off a preserved rock surface.

Associated sedimentary structures: (1) ‘old elephant skin’; (2) petees; (3) domal sand buildups.

Palaeoenvironment: intertidal to subtidal marine.

Ideas on genesis: The lack of any mud separating such sandstone veneers as well as variable OES textures on each of the underlying layers suggest that each thin sandy layer was resistant to erosion, loading, and overprinting by the succeeding thin sand veneer due to the presence of a biofilm upon each sandy layer.

Domal sand buildups: Figs. 4(a)-9 to -11

Name of structure: domal sand buildups.

Other terms used: ‘sand stromatolites’.

Description of structure: clustered and non-clustered low mound-shaped structures on upper sandstone bed surfaces; clustered ones may locally amalgamate. Cauliflower-like patterns form where clustered buildups occur, resembling carbonate stromatolites, but with much lower relief. Can be deformed by trace-making animals. Show no internal structure except for sandstones with relatively higher mud content, where convex laminae may be visible. Various sizes occur and these as well as clustered and non-clustered forms may occur on the same sandstone bedding surface; also preserved in concave hyporelief on the soles of overlying sandstone beds.

Associated sedimentary structures: (1) various trace fossils; (2) sand shadow structures; (3) oscillation and interference ripples; (4) 'Astropolithon'.

Palaeoenvironment: intertidal to shallow subtidal marine.

Ideas on genesis: it is thought that localized development of biofilms on sandy surfaces enable vertical accretion and/or binding of the quartz sand grains.

Sand shadow structures: Fig. 4(a)-12

Name of structure: sand shadow structures.

Other terms used: 'inverted flutes'.

Description of structure: these bed-top and bed-sole features resemble sand stromatolites in their size and shape, but additionally have a drumlin-like sand shadow accumulated in a downcurrent direction from the sand bed surface bumps. They lack any internal lamination, but have evidence of cohesiveness such as being cross-cut by trace-making animal trails.

Associated sedimentary structures: (1) various trace fossils; (2) sand stromatolites.

Palaeoenvironment: supratidal to shallow subtidal marine.

Ideas on genesis: microbial binding of sand may have produced these pliable yet cohesive surface bumps on sandy bed surfaces, with fallout of wind- or water-borne sand particles forming leeward tails behind these positive features.

Patchy ripples: Figs. 4(a)-13, -14

Name of structure: patchy ripples.

Other terms used: none.

Description of structure: flat sandstone surfaces in which patches are marked by ripples of various types, and where the ripples grade into the surrounding sandstone surfaces.

Associated sedimentary structures: (1) domal sand buildups; (2) various trace fossils; (3) 'Astropolithon'.

Palaeoenvironment: intertidal marine.

Ideas on genesis: ripples are able to form due to current, wave or wind action in patches within a mat-stabilised sandstone bed surface where the mat is either absent or has been removed by reworking processes (*sensu* Noffke et al., 1997b).

Cracked ripple crests: Fig. 4(a)-15

Name of structure: cracked ripple crests.

Other terms used: ripple-top sand cracks (see Fig. 4(c)-4B).

Description of structure: parallel sets of cracks along ripple crests in rippled sandstone bed surfaces.

Associated sedimentary structures: (1) sand stromatolites; (2) a range of mat-related features.

Palaeoenvironment: supratidal to shallow subtidal marine.

Ideas on genesis: biofilms are inferred to have covered sandy rippled surfaces and when these dewatered or even desiccated, the cracks formed preferentially along ripple crests where lateral stress may have been maximized.

Concave eroded ripple crests: Fig. 4(a)-16

Name of structure: concave eroded ripple crests.

Other terms used: broached ripples.

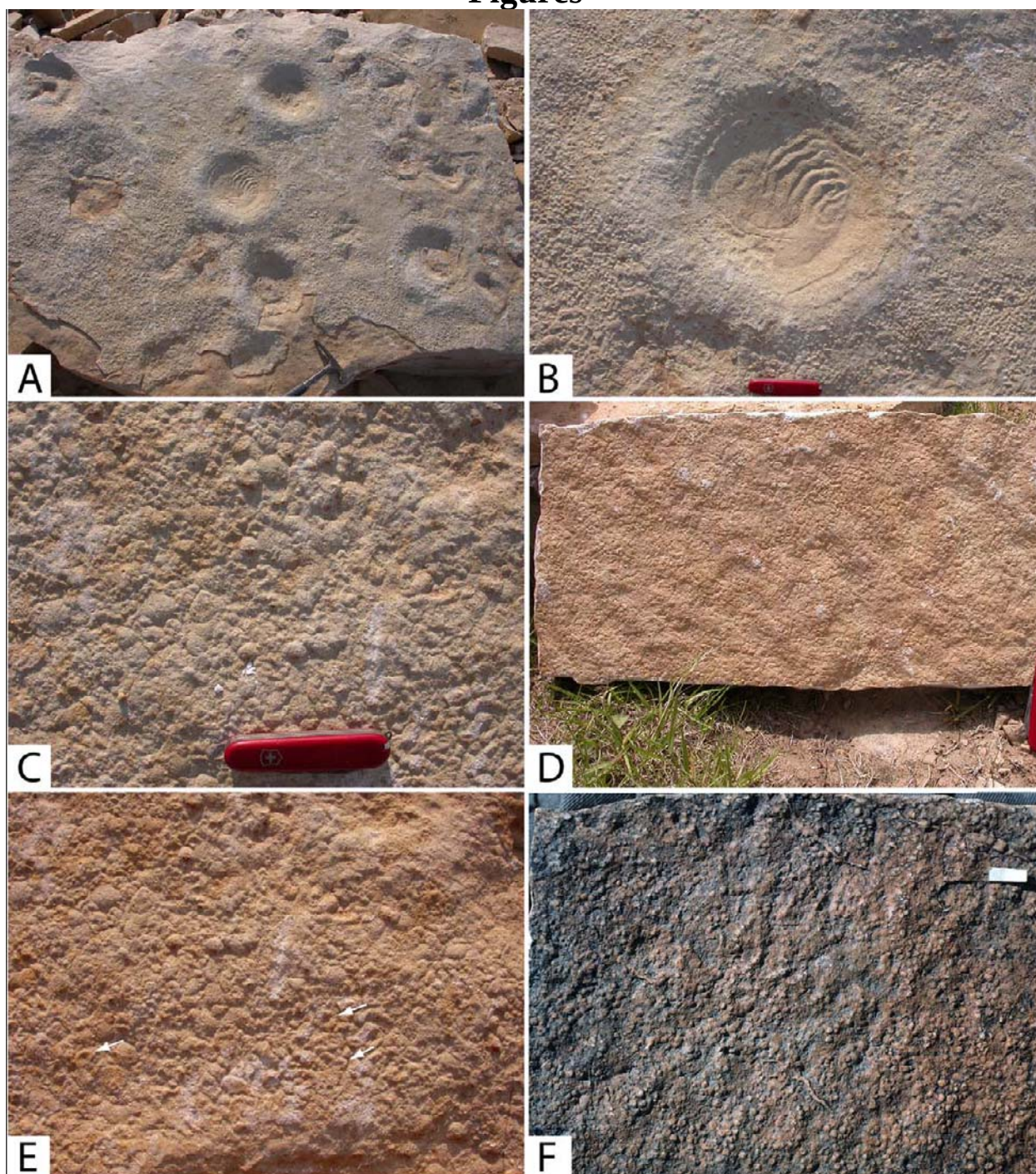
Description of structure: ripples are eroded in the sense that a shallow cavity is excavated in the ripple crests, rather than displaying either rounded or flat-topped crests that are the normal result of reworking of rippled sands. A distinct lip is also associated with these cavities.

Associated sedimentary structures: (1) ripple marks; (2) trace fossils.

Palaeoenvironment: intertidal to shallow subtidal marine.

Ideas on genesis: a rippled sandy surface stabilized by a microbial mat is envisaged. When high energy reworking processes remove both ripple crest mat and underlying loose sand, the rest of the ripple form remains relatively stable, enabling excavation of the shallow cavities in the crests, with formation of the lips due to microbial binding. May be genetically related to cracked ripple crests (above).

Figures



In: *Atlas of microbial mat features preserved within the clastic rock record*, Schieber, J., Bose, P.K., Eriksson, P.G., Banerjee, S., Sarkar, S., Altermann, W., and Catuneau, O., (Eds.) J. Schieber et al. (Eds.), Elsevier, p. 53-71. (2007)

Fig. 4(a)-1: Spheroidal pliable sand clasts, also known as ‘algal balls’, ‘sand balls’, or ‘sand ooids’, in quartz arenites of the Late Cambrian Elk Mound Group, Wisconsin, USA (A-E), and of the Cambrian Fish River Formation, Namibia (F):

Images A-E are bed-soles; A-B bear scyphomedusae impressions. All surfaces are overlain and underlain by very fine- to medium-grained quartz arenites. (B) is a close-up of the centre of (A), and (C) is a close-up of the lower right portion of B. (E) is a close-up of the upper left portion of (D). Spheroidal clasts in A-C are preserved in convex hyporelief (figured) and convex epirelief (not figured; but on reverse side of exfoliating sand veneer near hammer in A). Note the deflated/compressed clasts in E, some of which are arrowed. Slab, X-ray, and petrographic examination of serially sectioned spheroidal sand clasts in A-C revealed no concentric internal lamination, clay, or evidence for non-quartzose cements. Together with the fact that these clasts deform one another, yet are not preferentially sorted into ripple troughs or topographic depressions, these properties suggest that they were likely pliable and possibly also sticky - properties consistent with fecal pellets or microbially bound sand spheres. Pliability of clasts and deformation of overlying and underlying beds (implying that all three were unlithified) diminishes the possibility that these clasts could be weathered beach rock fragments. Specimen in F is from P. Pflüger, Tübingen University. Hammer head in A is 17.5 cm long; Swiss army knife in B-D is 8.3 cm long; scale in F is 1 cm long. All photos: J.W. Hagadorn.

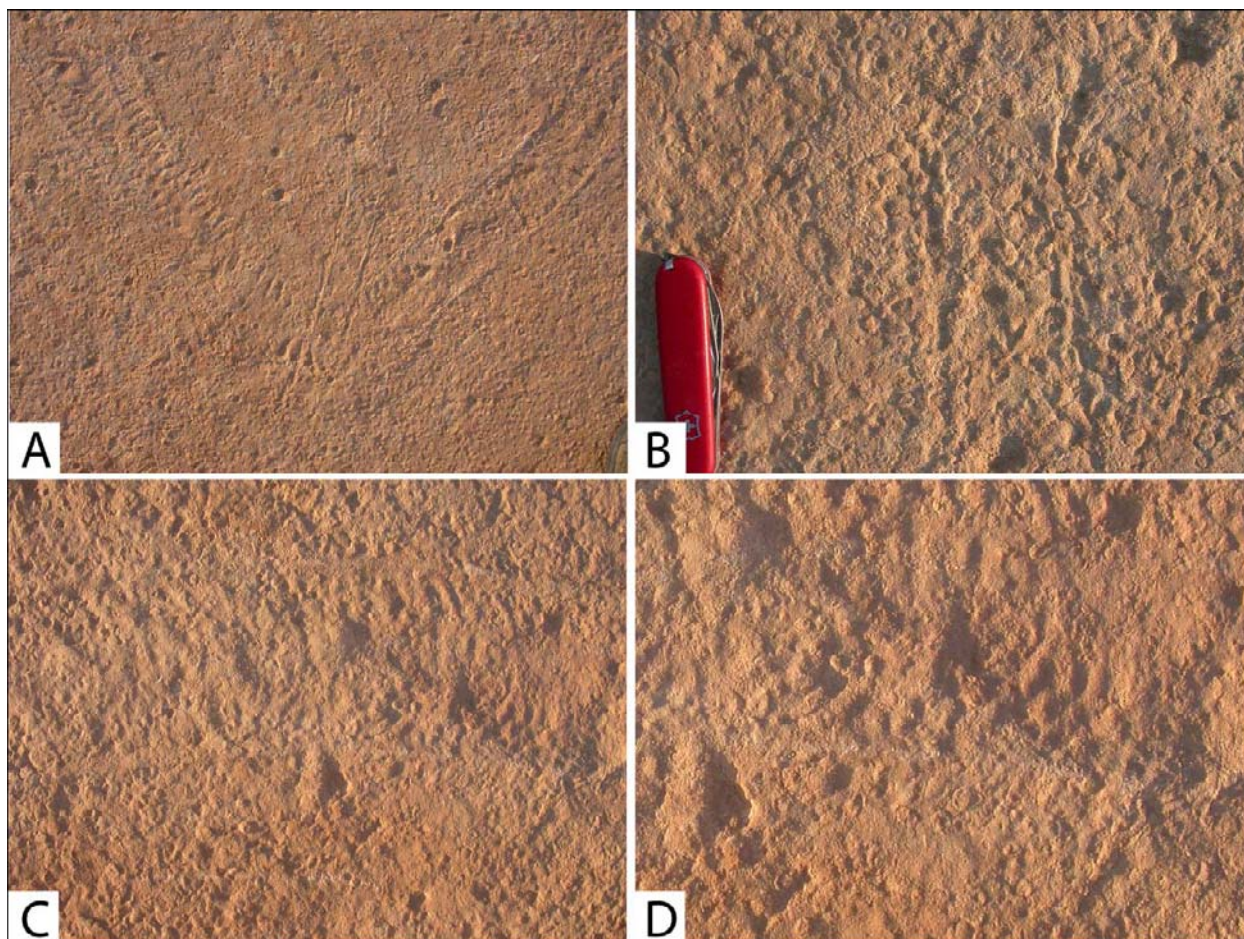


Fig. 4(a)-2: Spheroidal pliable sand clasts in quartz arenites of the Late Cambrian Elk Mound Group, Wisconsin, USA:

All images are of the same bed surface, that bears *Climactichnites* (A, C, D) and *Diplichnites*-like (A, B) trackways, which have pliable sand clasts superimposed atop and impressed into them. (B) is a close-up of the centre of (A), and (D) is a close-up of the right-centre portion of (C). In B and D, note where trace fossils have been plially deformed by the sand clasts, and that where clasts have been removed, crater-like features remain. Exfoliation of the bed overlying this surface reveals that spheroidal sand clasts are sandwiched between medium-grained quartz arenite beds, but create convex impressions in both overlying and underlying beds. Note that clasts exhibit evidence of folding, mutual deformation, and partial deflation. In petrographic and X-radiographic analyses of thin-sectioned clasts, no concentric internal lamination, clay, or evidence for non-quartzose cements is present. *Climactichnites* trails in A, C, D are 12 cm wide, Swiss army knife in B is 8.3 cm long. All photos: J.W. Hagadorn.

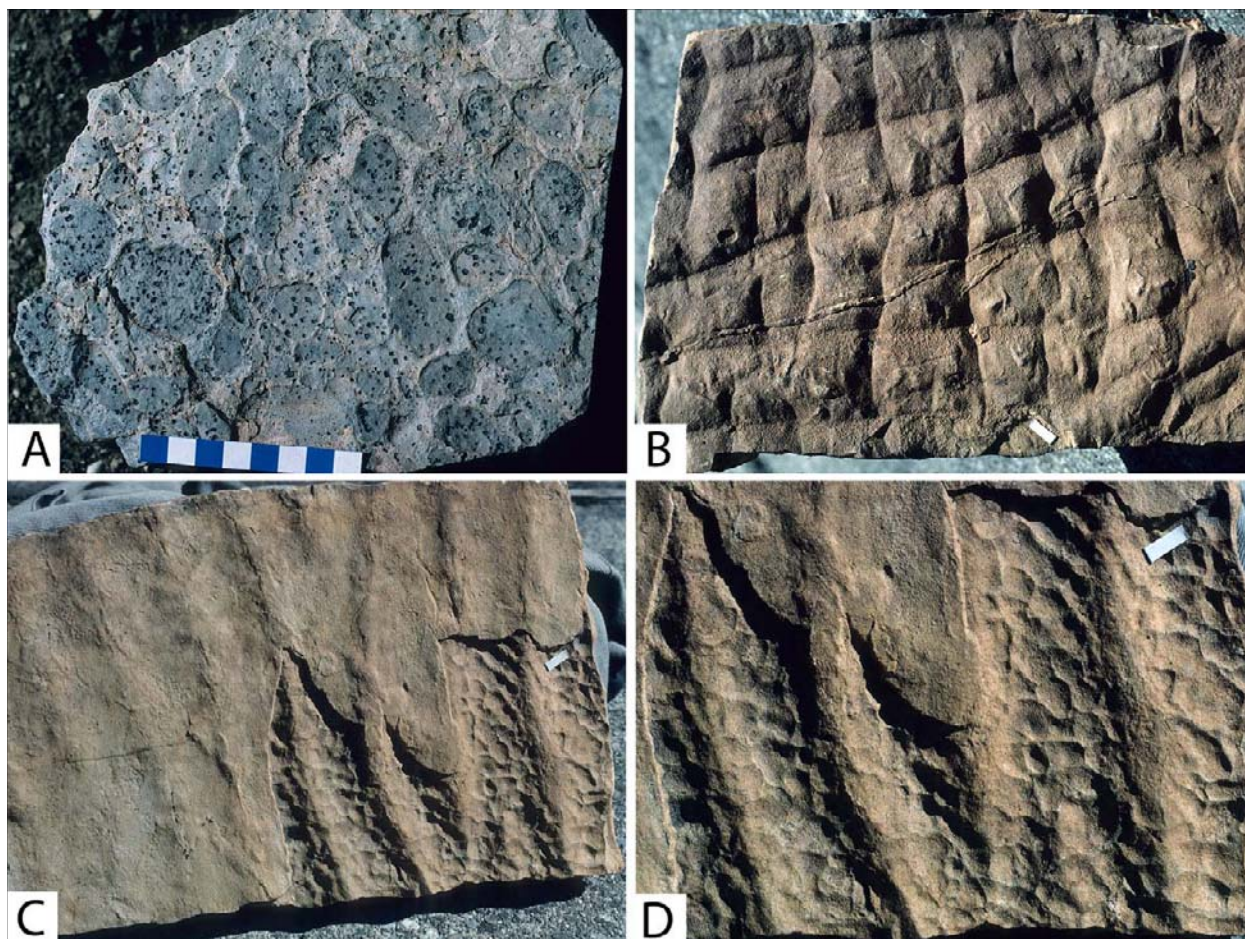
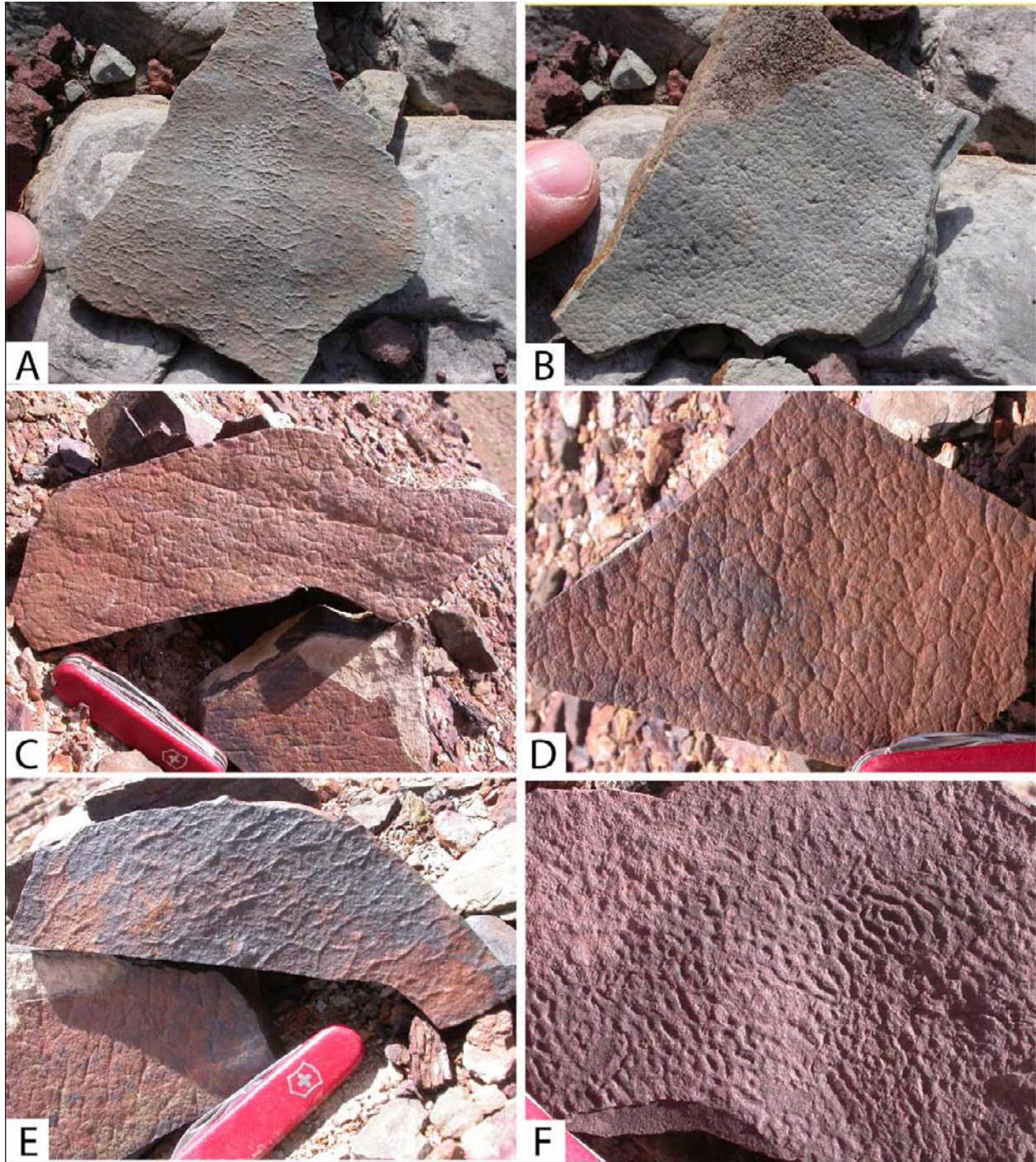


Fig. 4(a-3): Sand chips (A), palimpsest ripples (B) and loaded ripples (C-D):

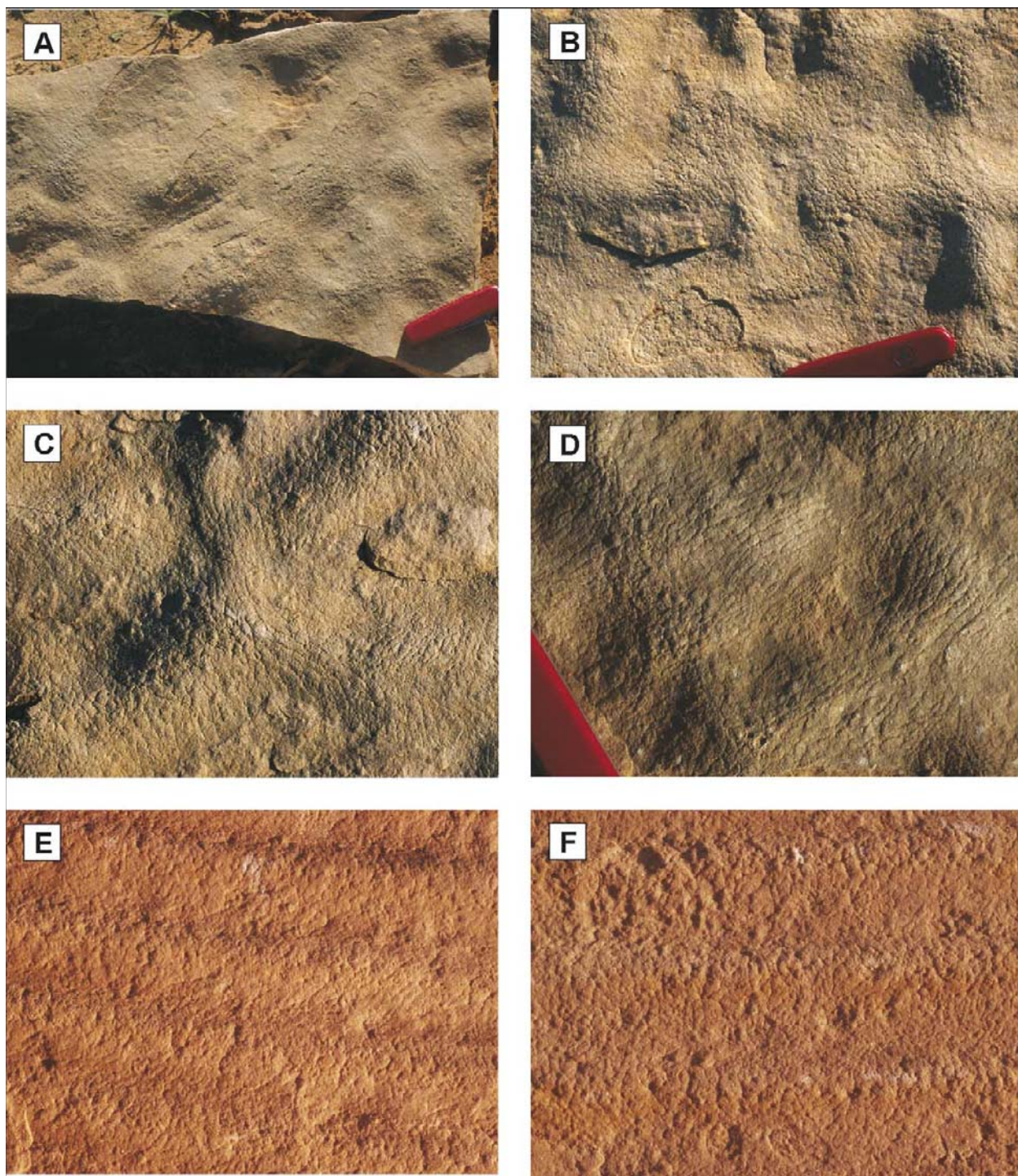
(A) is from the Neoproterozoic Nudaus Formation, Namibia (Pflüger and Gresse, 1996). (B-D) are from the Mesoproterozoic Chorhat Sandstone, India. A, C, D, are bed surfaces, B is a bed-sole. Note preferential pyritization of quartz arenite sand chips in A, and preservation of older ripple crests and younger ripple crests in B. Both the rounding and transport of unlithified quartz sand in A, and the lack of erosional scouring of older sharp ripple crests (by the younger set) in B suggest that organically-mediated sand stabilization occurred - possibly resulting from microbial binding of older surfaces. In C and D, note the loading of oscillation ripple-marked sand by a thin veneer of overlying sand. The absence of clay at this bed interface, together with the absence of amalgamation of the two lithologically identical layers suggests that there must have existed some type of biofilm or other veneer which inhibited amalgamation, but was flexible enough to allow plastic deformation at the time of burial of the underlying rippled surface. Specimens from P. Pflüger, Tübingen University. Scale bar increments in A are 1 cm, and bars in B-D are 1 cm long. All photos: J.W. Hagadorn.



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Fig. 4(a)-4: Wrinkle structures, including ‘runzelmarken’/wrinkle marks (A-E) and ‘Kinneyia’ (D) in fine-grained sandstones. A, B: Carbon Canyon Member, and C-E, Jupiter Member, Neoproterozoic Chuar Group, Grand Canyon, USA. F: Neoproterozoic Nankoweap Formation, Grand Canyon, USA:

(A, B) Freshly split fragments of part-counterpart wrinkle-bearing surfaces; upper surface at left. (C, D) Bed-soles of oxidized (limonitic?) wrinkled surfaces; counterpart (bed surface) to this horizon is in (E). Note raised reticulate pattern on bed tops in A, E, and bed-soles that display what look like small-scale loading features or ‘Old Elephant Skin’ (OES) in B-D (see also Figs. 4(a)-5 to -8 for additional examples of OES). Wrinkle structures are poorly understood, but in sandstones in which mud is absent between sandstone bed interfaces, may be formed by purely microbial processes or a combination of biofilm-mediated physical loading processes. (F) ‘Kinneyia’, characterized by an irregular mosaic-pattern of flat-topped microripple-like crests. Such structures originally were thought to be biogenic (*sensu Kinneyia*), were then assigned to physically produced structures, and have recently been found to form through gas bubble buildup inhibiting cementation beneath a sandy mat (Pflüger, 1999). Thumbnail in A-B is ~2 cm wide; Swiss army knife in C-F is 8.3 cm long. All photos: J.W. Hagadorn.



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Fig. 4(a)-5: ‘Old elephant skin’ , also known as OES, a type of wrinkle structure visible on bed-soles at sand-sand bed interfaces; all images are of quartz arenites of the Late Cambrian Elk Mound Group, Wisconsin, USA:

This quasi-polygonal texture may form when biofilms provide a veneer between beds upon which loading occurs, yet in which the biofilm keeps the overlying and underlying beds (which are lithologically identical) from amalgamating. Modern mats have also been observed to produce a similar surface corrugation, and sand deposited atop such mats may cast this mat-produced surface sculpture. Many OES-bearing surfaces are also characterized by exfoliating sand laminae (compare A, B to Figs. 4(a)-6A, -6D, 4(a)-8). Given the very thin nature of overlying sand laminae (visible at bottom of B), where <0.5 mm thick quartz laminae are superimposed on top of one another, yet each have different and non-overlapping surface patterns, it seems plausible that this texture represents a primary surface-produced feature. (B-D) Close-up views of the ripple-marked bed-sole in (A), viewed with different lighting. (E-F) are part-counterpart slabs of a straight-crested oscillation ripple-marked bed interface. Swiss army knife in A, B, D is 8.3 cm long, scale bars in C, E, F are ~3 cm long. All photos: J.W. Hagadorn.

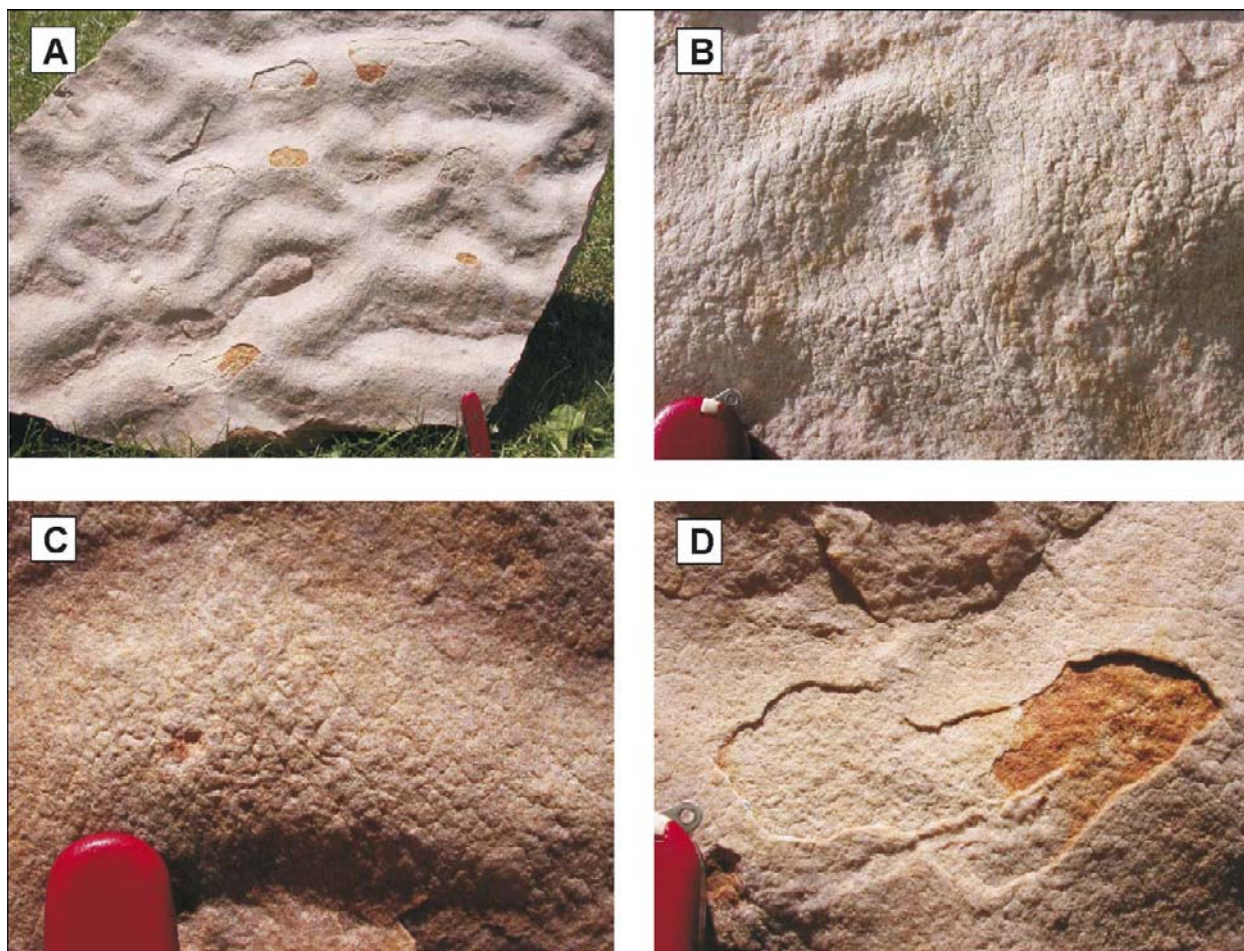
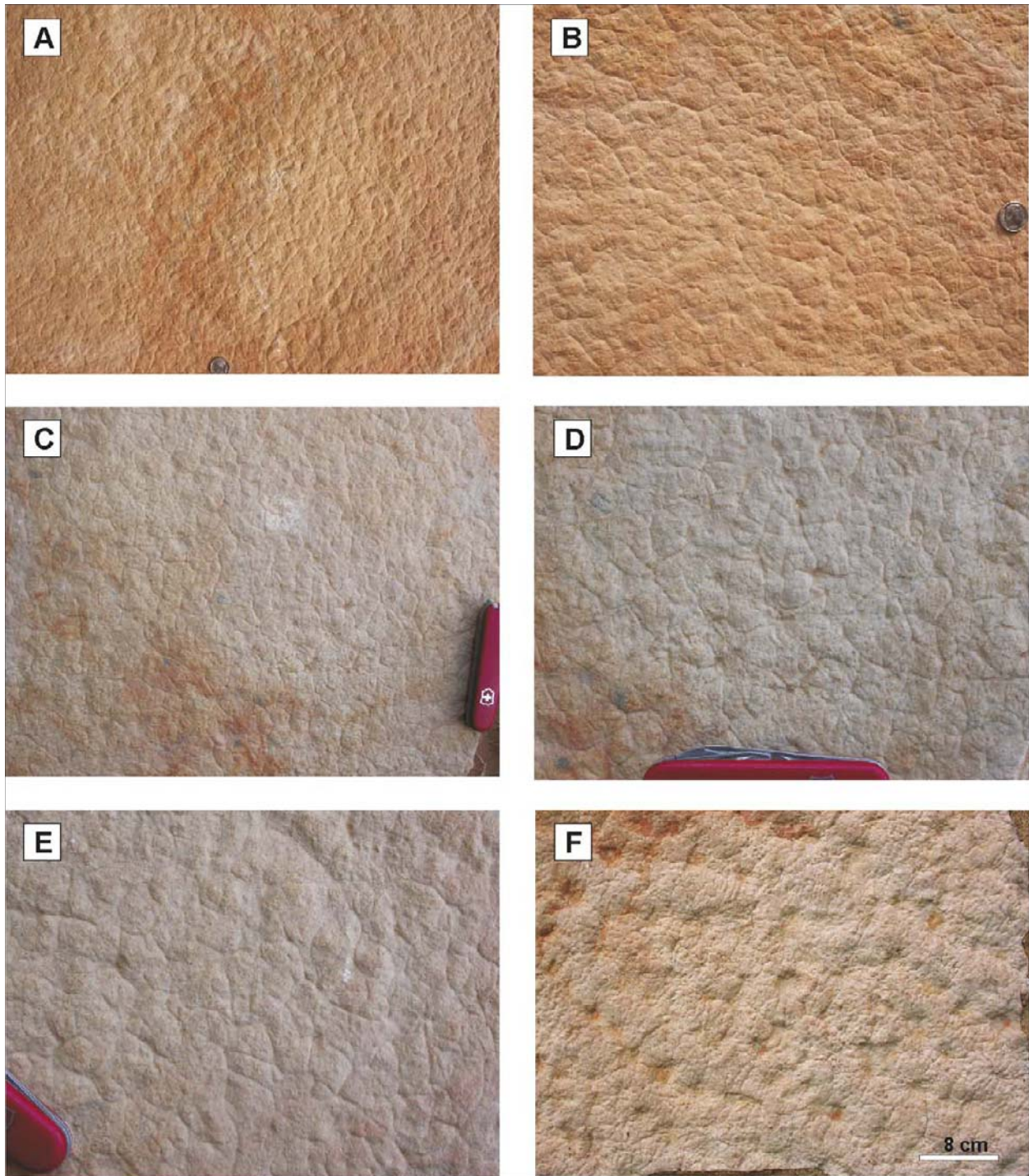


Fig. 4(a)-6: ‘Old elephant skin’ , or OES, on bed-soles in quartz arenites of the Late Cambrian Elk Mound Group, Wisconsin, USA:

This quasi-polygonal texture may form when biofilms provide a veneer between beds upon which loading processes can operate, yet in which the biofilm kept the overlying and underlying beds (which are lithologically identical) from amalgamating. On many freshly parted surfaces, an oxidized veneer or limonitic coating is visible at or between sand-sand bed interfaces (e.g., orange-coloured regions visible in A, D, where underlying layer has been removed; for comparison see freshly removed areas between sand laminae in Fig. 4(a)-8). Gehling (1999) hypothesized that this coating represents oxidized remnants of bacterially mediated pyrite precipitated in/around a decomposing biofilm. (B-D) These are close-ups of the ripple-marked bed-sole in (A), viewed with different lighting. Swiss army knife in A-D is 8.3 cm long. All photos: J.W. Hagadorn.



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Fig. 4(a)-7: Large OES ('old elephant skin') textures, a type of wrinkle structure visible on bed-soles at sand-sand bed interfaces. A-E are from quartz arenites of the Late Cambrian Elk Mound Group, Wisconsin, and F is from the Late Cambrian Gunter Sandstone, Missouri, USA:

Unlike smaller-sized OES, this type of large 0.5-2 cm scale wrinkle structure is not well known, and is not always associated with exfoliation of thin sand laminae; investigations of how it is manifested in vertical section have not been conducted. (B) is close-up of (A), and (D-E) are close-ups of (C), lit from different angles. Note wicking and quasi-radial patterns to OES forming on ripple crests and troughs in (F). Coin in A-B is 1.8 cm wide; Swiss army knife in C-E is 8.3 cm long; scale in F is 8 cm long. All photos: J.W. Hagadorn.

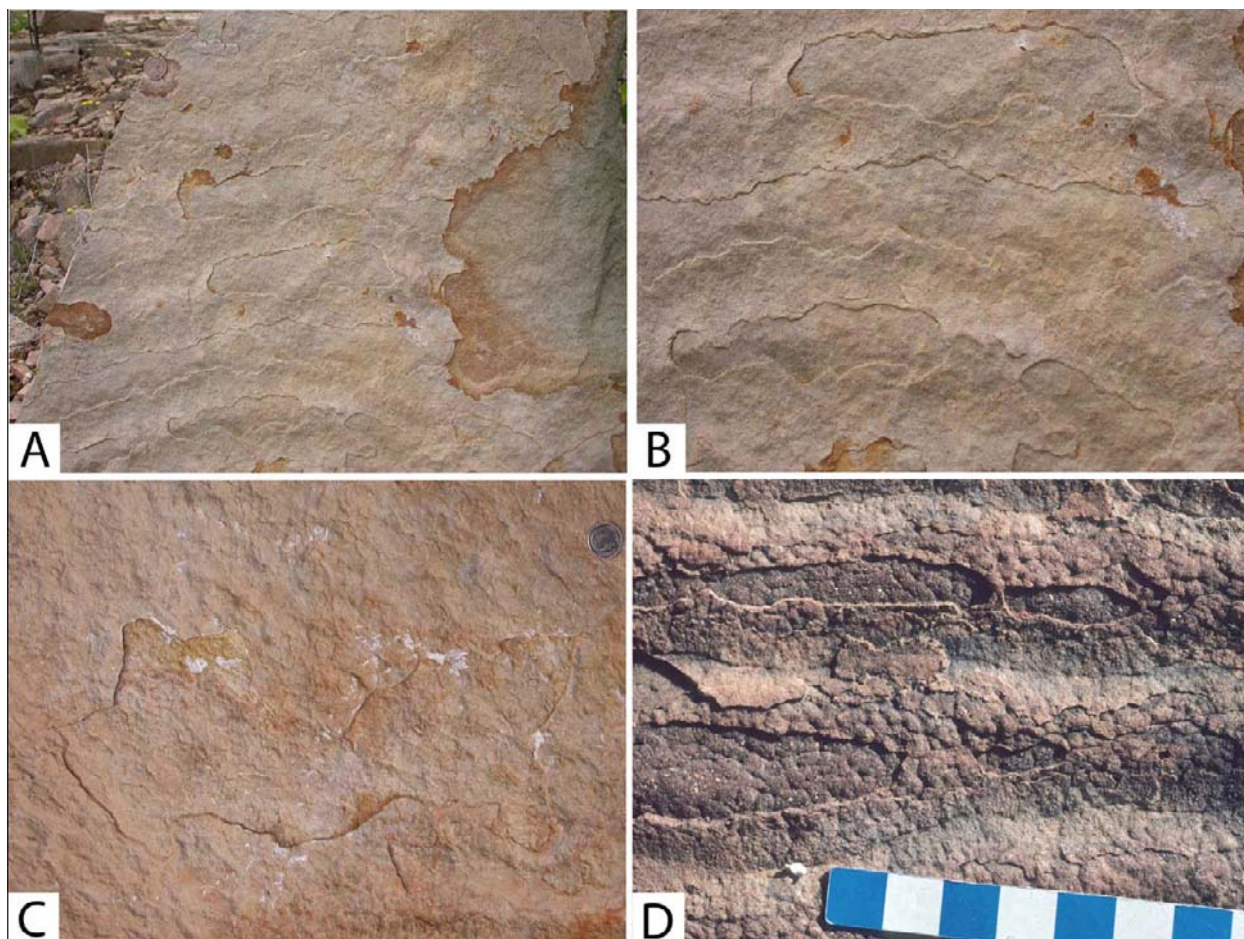
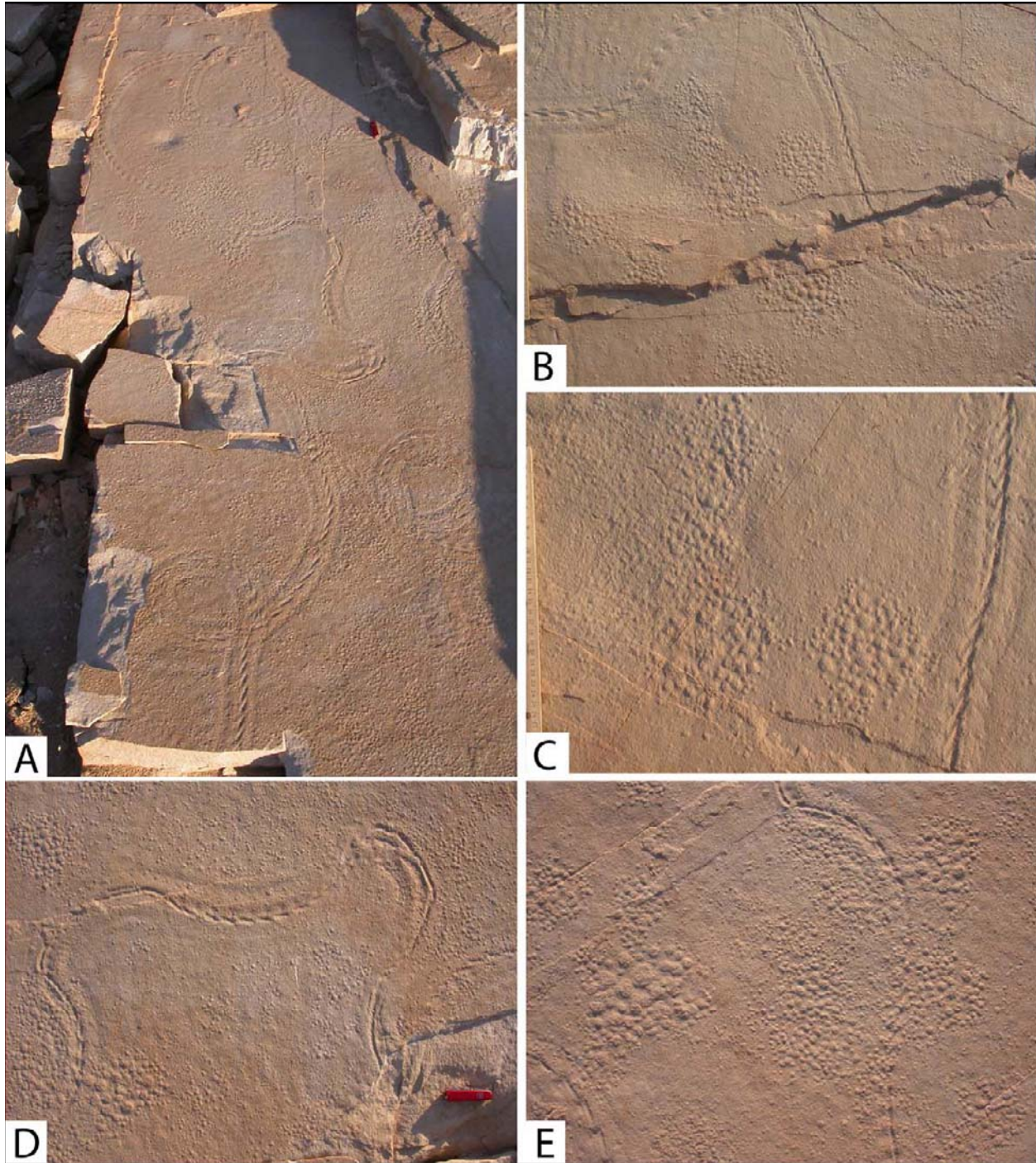


Fig. 4(a)-8: Exfoliating sand laminae visible on bed-soles of rippled quartz arenites. A-C are from the Late Cambrian Elk Mound Group, Wisconsin, USA; D is from the Neoproterozoic Rawnsley Quartzite, South Australia:

(B) is a close-up of the central area in (A). Note the successive ≤ 1 mm thick veneers of medium-grained sand spalling off the rock surface; no mud occurs between these laminae, and reticulate elephant skin structure occurring in overlying layers is not superimposed on or through underlying layers. Thus, wrinkled OES textures on each underlying layer differ in their arrangement from those of the overlying layer. Together, these features suggest that each veneer was resistant to erosion, loading, and imprinting by the overlying burial layer - and did not become amalgamated into or adhere to the compositionally identical overlying non-muddy layer. Such behaviour is consistent with the presence of a biofilm on each layer, which inhibited amalgamation of the underlying layer into the newly deposited overlying layer. A limonite coating is often visible between the two layers (orange areas where laminae have been freshly removed), and could result from oxidation of bacterially mediated pyrite beneath each biomat (*sensu* Gehling, 1999). Coin in A, C is 1.8 cm wide; scale bar units in D are cm. All photos: J.W. Hagadorn.



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Fig. 4(a)-9: Domal sand buildups, also known as ‘sand stromatolites’ on a bed surface characterized by *Protichnites* trackways in quartz arenites of the Late Cambrian Elk Mound Group, Wisconsin, USA:

These low mound-like structures often occur in clusters, with larger heads toward their centre and smaller heads toward their exterior. Similar cauliflower-like growth morphologies are exhibited by stromatolites in carbonate settings, except that the sand stromatolite heads have significantly lower vertical relief - probably owing to the lack of bacterially mediated carbonate precipitation in heads - the latter normally facilitates upward growth of heads and physical resistivity to waves, wind, and tides. Images (B-E) are close-ups of surface in (A), and E is a close-up of left portion of D, where track cross-cuts or is grown-over/-through by sand stromatolite clusters. Swiss army knife in A, D is 8.3 cm long. All photos: J.W. Hagadorn.

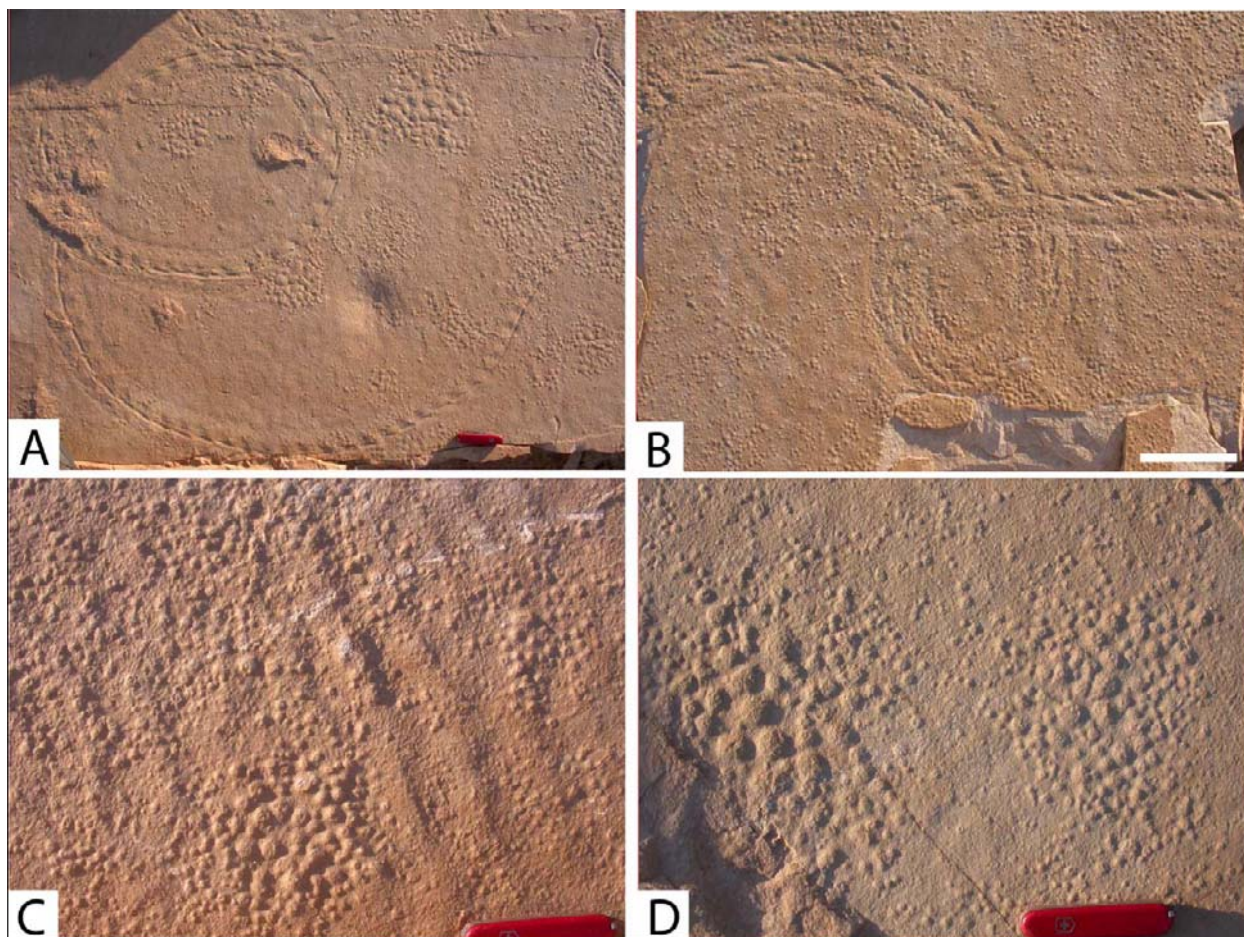
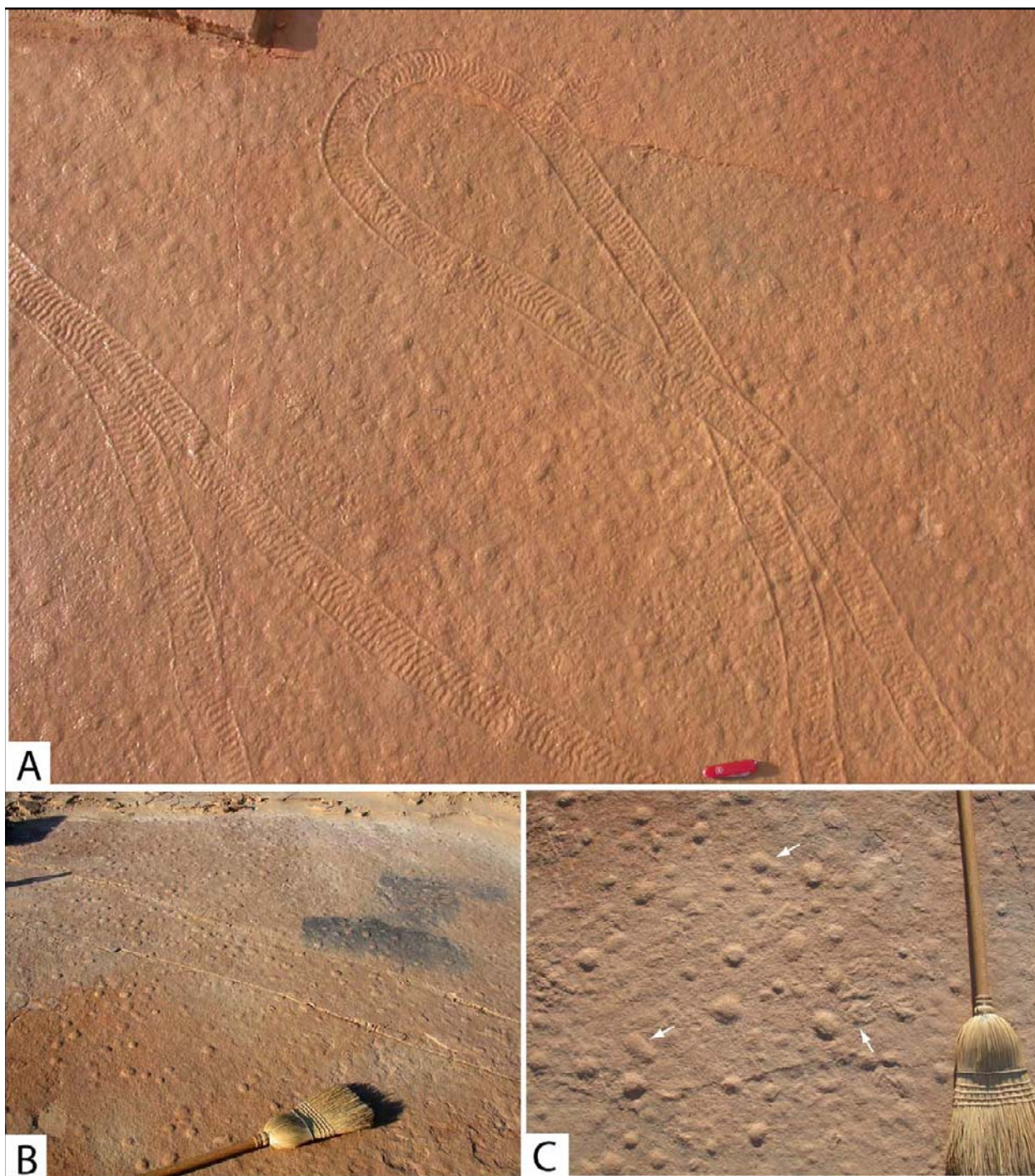


Fig. 4(a)-10: Domal sand buildups, also known as ‘sand stromatolites’ on quartz arenite bed surfaces of the Late Cambrian Elk Mound Group, Wisconsin, USA:

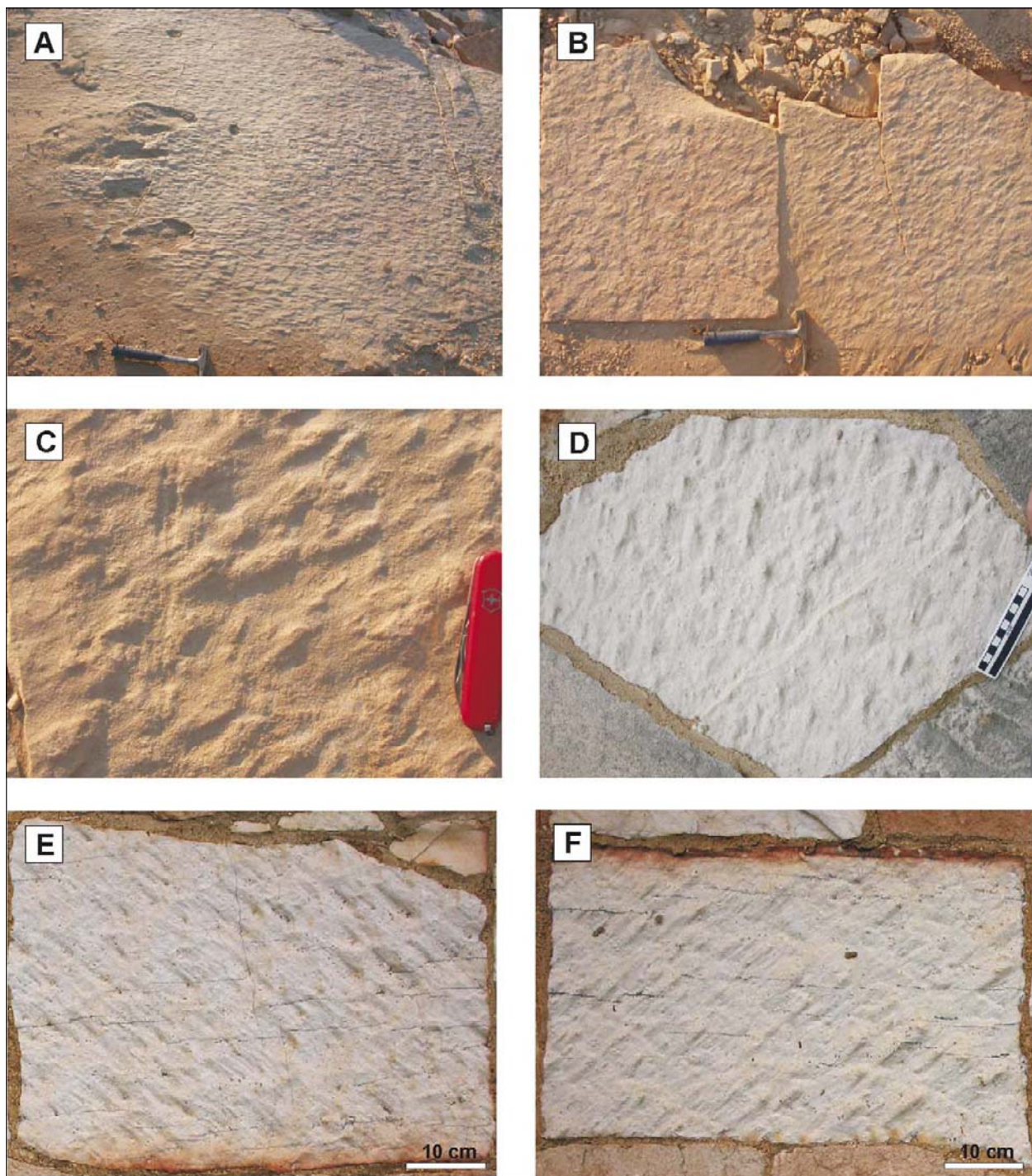
(A) and (B) are from the same surface, and are characterized by looping *Protichnites* trackways. The area shown in A has well clustered heads, yet B is dominated by non-clustered small pea-sized sand stromatolites; together these indicate that both forms may exist under the same environmental conditions. The different morphologies thus may represent different life histories, biological affinities, and/or histories of interaction with the physical environment. Clustered sand stromatolites can also grow over rippled (C) and unrippled (D) surfaces, and together with the mixture of unclustered and clustered forms in C, suggest that perhaps clustered sand stromatolites may represent an older or faster-growing region than the unclustered heads. Although it is possible that large sand stromatolites (Fig. 4(a)-11) represent the oldest, fastest-growing, and/or most resistant components of such microbial communities, it is unclear how one could confidently falsify or constrain these hypotheses. Swiss army knife in A, C, D is 8.3 cm long; scale in B is ~12 cm long. All photos: J.W. Hagadorn.



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Fig. 4(a)-11: Large non-clustered domal sand buildups, also known as ‘sand stromatolites’, on quartz arenite bed surfaces of the Late Cambrian Elk Mound Group, Wisconsin, USA:

These heads are relatively evenly distributed on these surfaces, and range from 0.1-2.5 cm high and 0.2-10 cm wide. Occasionally heads have grown/merged together (arrowed in C), but unlike clustered forms, smaller sand stromatolites do not occur on the periphery of larger heads. Upward-oriented arrow indicates where the tops of two joined stromatolite heads have been removed, revealing the amalgamation of two larger heads into one elongate barbell-shaped head. Structures exhibit evidence of having synoptic relief (note sand shadows in Fig. 4(a)-12), yet are soft enough to be deformed by trace-making animals (e.g., the trackways that cross over and cross-cut domes in A and in Figs. 4(a)-9, -10, -12). Study of cross-cutting relationships on the 200 m² surface in (A) revealed that sand stromatolites grew both before and after production of the surface-produced *Climactichnites* traces. Together with analyses of part-counterpart specimens that falsify an inorganic origin (e.g., as a concretion, nodule, sub-mat gas bubble) for all of the surfaces shown in section 4(a), this information suggests that these structures might form by the vertical accretion and/or binding of quartz sand by localized biofilms. Serial slab, X-ray CT, and petrographic thin-section analyses of these structures reveal no internal structure, although in muddier sands (*sensu* the Cambro-Ordovician Nepean Formation sand stromatolites – see Chapter 8(b)), weak domal laminations are visible in cross-sectioned sand stromatolites. (C) is close-up of (B). Swiss army knife in A is 8.3 cm long, broom handle is 4 cm wide in B-C. All photos: J.W. Hagadorn.



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Fig. 4(a)-12: Sand shadow structures, also known as ‘inverted flutes’, in quartz arenites of the Late Cambrian Elk Mound Group, Wisconsin (A-C), and Gunter Sandstone, Missouri (D-F), USA:

Images A-E are bed-tops; F is a bed-sole and is the counterpart to E. (A) illustrates a surface pavement in which sand has accumulated in the downcurrent direction of surface bumps, forming drumlin-shaped sand shadows. Although serial sectioning of part-counterpart slabs from this surface reveals no internal lamination in the upcurrent end of the bumps, these bumps exhibit deformational features suggesting that they were cohesive, yet pliable - unlike nodules or concretions. For example, in the close-up of this surface (B, C) the stoss and leeward portions of sand stromatolite-like surface bumps are commonly cross-cut by *Cruziana* (arrowed in B, *Gordia*, *Helminthoidichnites*, and *Protichnites*), indicating that at the time of deposition, the core of the sand shadow was still pliable. Sand bumps are similar in size and shape to sand stromatolite heads, and are not expressed below bed surfaces. Given the absence of clay in these lithofacies, microbial binding provides one explanation for the cohesive, yet pliable rheology which mediated formation of sand bumps, followed by sand shadows. Fallout of wind- or water-borne grains led to buildup of leeward tails of sediment behind sand bumps. Hammer head in A, B is 17.5 cm long; Swiss army knife in C is 8.3 cm long; scales in (D-F) are 10 cm long. All photos: J.W. Hagadorn.



Fig. 4(a)-13: Patchy ripples, on oscillation rippled quartz arenite from the Upper Cambrian Elk Mound Group, Wisconsin, USA:

See close-ups of this surface in Figure 4(a)-14. Because the surface is nearly perfectly flat, the hypothesis that rippled patches could be forming on topographic highs within a tide-pool can be falsified. Instead, ripples are thought to form in regions where biofilms and biomats have either been ripped up or were not present (*sensu* Noffke et al., 1997b). In areas where ripples are absent, abundant sand stromatolite heads are present. Note also the '*Astropolithon*' (A), a probable sub-mat gas blister structure, in the zone characterized by unrippled sand stromatolite-laden sand. Also note the *Teichichnus*-like trace fossils (arrowed) on the surface - these cross-cut both ripple patches and sand stromatolite-laden patches, demonstrating that the sediment was pliable enough to be burrowed. Dark areas at top of photo are wet. Field of view is ~3 m. Photo: J.W. Hagadorn.

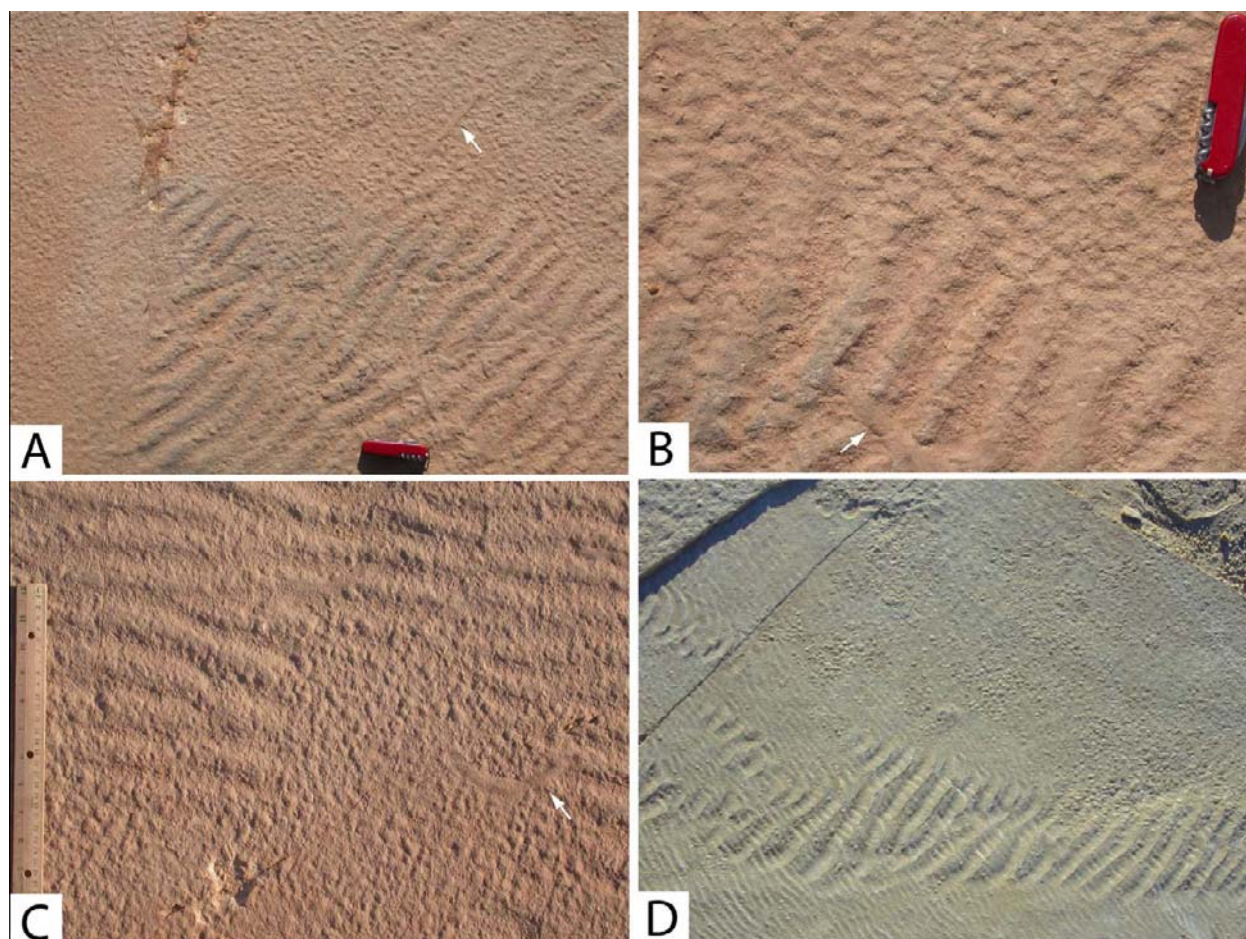


Fig. 4(a)-14: Patchy ripples from the Late Cambrian Elk Mound Group, Wisconsin, USA:

Surfaces in (A-C) are close-ups of surface in Figure 4(a)-13. Note pustulose, sand stromatolite-laden nature of unrippled regions in A-C. Some of the ripple crests and troughs in C, and on another surface in (D) have been colonized by biofilms after ripple formation, suggesting that at least two generations of microbial development occurred prior to burial. Isolated horizontal trace fossils are arrowed; note cluster of burrows above knife in A. The surface in D is similar to A-C, but is characterized by patches of clustered sand stromatolites in non-rippled regions. The second set of ripples on this surface is migrating over the top of the first ripple set and the sand stromatolite-covered region; crests of the second set are oriented from the upper left to lower right of the photo. Swiss army knife in A, B is 8.3 cm long; ruler in C is 31 cm long; field of view in D is 2 m. All photos: J.W. Hagadorn.

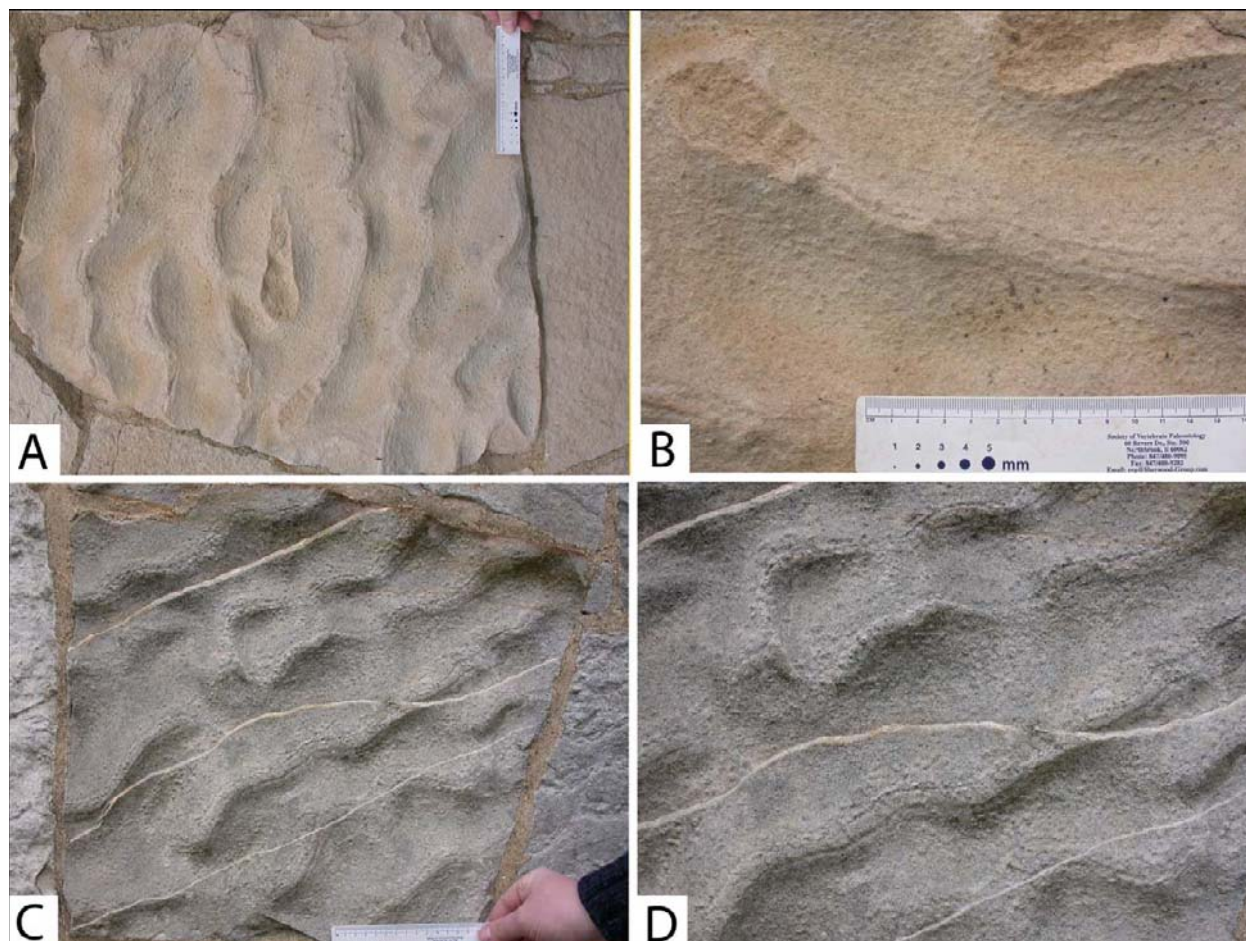


Fig. 4(a)-15: Cracked ripple crests, with incipient sand stromatolites coating ripples in quartz arenites from the Late Cambrian Gunter Sandstone, Missouri, USA:

(B) and (D) are close-ups of (A) and (C), respectively. These structures are poorly understood, but examination of part-counterpart specimens and their occurrence at sand-sand bed interfaces suggests that linear ridge-crest features could form when exposed tops of organically-bound ripples crack when dewatering. Whether this occurs subaerially or subaqueously is unknown. Abundant small (incipient) sand stromatolites are visible in D, and seem to be concentrated on ripple crests. The presence of microbial indicators elsewhere on these surfaces is consistent with the interpretation that biofilms are involved in the production of parallel sets of cracks along ripple crests, perhaps due to dehydration of bound sand. Although cracks occur at sand-sand bed interfaces, serially sectioned specimens have not been examined, and thus the origin and subsurface characteristics of these structures are still not well constrained. See also Figure 4(c)-4B, this chapter. Scale bar units are 1 cm. All photos: J.W. Hagadorn.

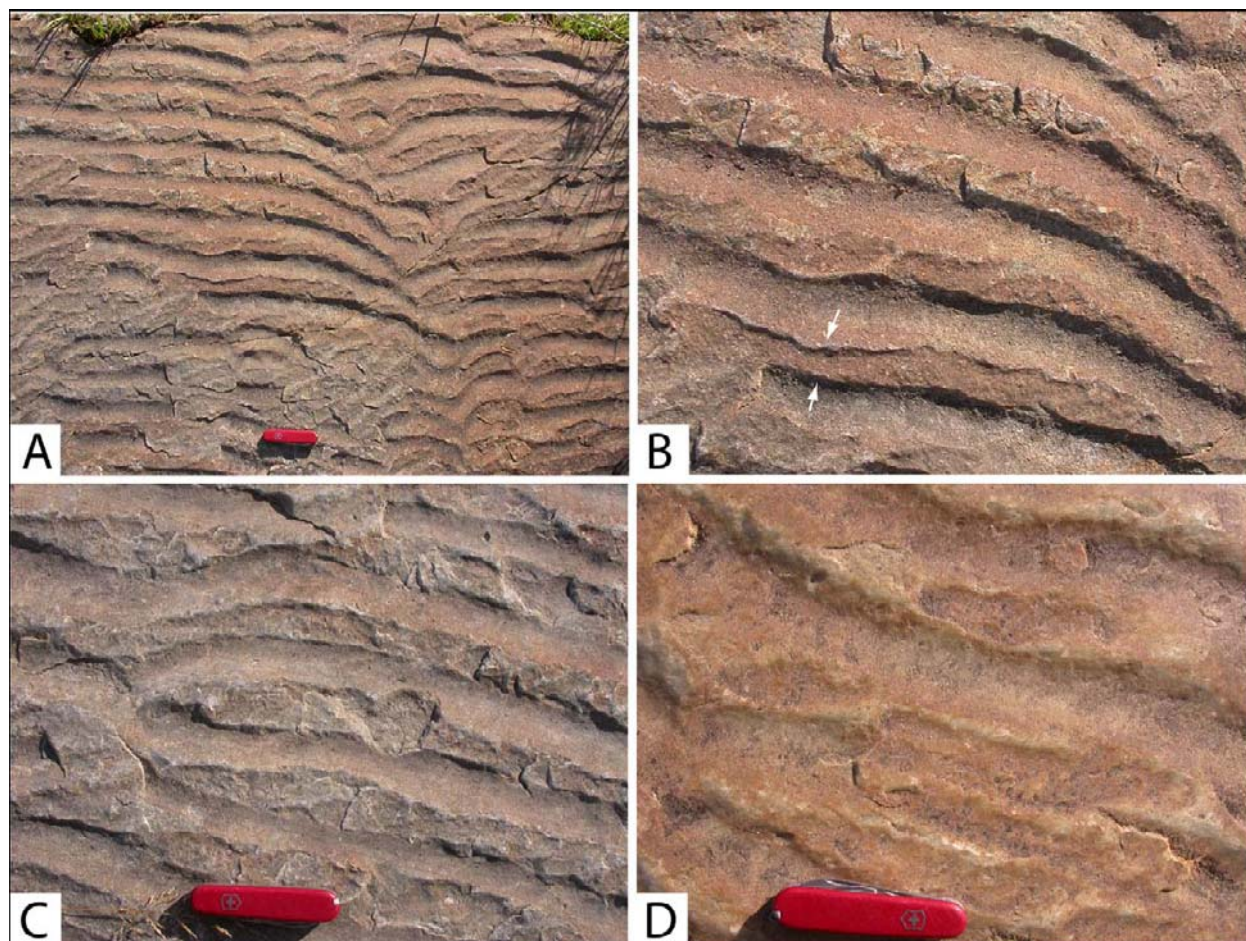


Fig. 4(a)-16: Concave eroded ripple crests on bed surfaces of quartz arenites in the Late Cambrian Potsdam Group, New York, USA:

(B) is a close-up of central area in (A). Prior to or concomitant with burial of this layer by the veneer of sand above the pocket knife, the ripple crests were eroded and reworked. However, rather than forming flat-topped ripple crests, crests are excavated deeper than the adjacent stoss/lee sides of ripples, forming in some cases a cavity beneath the stoss and lee ripple surfaces. The margins of these cavities have a distinct lip (see close-ups in B and (C); and arrowed example in B), as if the stoss and lee surfaces of the ripples were cohesive or more resistant to erosion during the erosive phase of reworking. One possible explanation for this phenomenon is that the surface of ripples had been stabilized prior to being buried and having their tops scoured. After their crests were plucked off, turbulent flow regimes could entrain unbound grains in the ripple crest, including grains that lay adjacent to and obliquely underneath a biofilm veneer that coated ripple surfaces. Similar features have been noted in Neoproterozoic sandstones by Pflüger (1999). However, diagenetic or other predepositional scenarios to account for these unusual ripple crest cavities are also possible, and sectioning is needed to evaluate all of these hypotheses. Swiss army knife is 8.3 cm long. All photos: J.W. Hagadorn.

References

- Gehling, J.G., 1999. Microbial mats in terminal Proterozoic siliciclastics: Ediacaran death masks. *Palaios* 14: 40-57.
- Gerdes, G., Klenke, T., Noffke, N., 2000a. Microbial signatures in peritidal siliciclastic sediments: a catalogue. *Sedimentology* 47: 279-308.
- Hagadorn, J.W., Bottjer, D.J., 1997. Wrinkle structures: microbially mediated sedimentary structures common in subtidal siliciclastic settings at the Proterozoic-Phanerozoic transition. *Geology* 25: 1047-1050.
- Hagadorn, J.W., Bottjer, D.J., 1999. Restriction of a Late Neoproterozoic biotope: suspect-microbial structures and trace fossils at the Vendian-Cambrian transition. *Palaios* 14: 73-85.
- Noffke, N., Gerdes, G., Klenke, T., Krumbein, W.E., 1997b. Biofilm impact on sedimentary structures in siliciclastic tidal flats. *Courier Forschungsinstitut Senckenberg* 201: 297-305.
- Noffke, N., Gerdes, G., Klenke, T., Krumbein, W.E., 2001a. Microbially induced sedimentary structures - a new category within the classification of primary sedimentary structures. *J. Sediment. Res.* A71: 649-656.
- Pflüger, F., 1999. Matground structures and redox facies. *Palaios* 14: 25-39.
- Pflüger, F., Gresse, P.G., 1996. Microbial sand chips- a non-actualistic sedimentary structure. *Sediment. Geol.* 102: 263-274.
- Schieber, J., 2004. Microbial mats in the siliciclastic rock record: a summary of the diagnostic features. In: Eriksson, P.G., Altermann, W., Nelson, D.R., Mueller, W.U., Catuneanu, O. (Eds.), *The Precambrian Earth: Tempos and Events. Developments in Precambrian Geology* 12, Elsevier, Amsterdam, pp. 663-673.
- In: *Atlas of microbial mat features preserved within the clastic rock record*, Schieber, J., Bose, P.K., Eriksson, P.G., Banerjee, S., Sarkar, S., Altermann, W., and Catuneanu, O., (Eds.) J. Schieber et al. (Eds.), Elsevier, p. 53-71. (2007)